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Data Analysis Tasks – CodeAlpha

This project aims to explore smartphone addiction using exploratory data analysis (EDA) and Tableau visualizations.

Tools Used;

- Microsoft SQL Server Management Studio
- Python
- Microsoft Excel
- Tableau

Task 2 – Exploratory Data Analysis (EDA)

Some of the questions that I found asking myself when working on this dataset included;

- Which addiction level criteria has the highest average screentime?
- Does social media usage aid to smartphone addiction than work/study?
- What is the relationship between gaming hours and addiction level
- What is the average notification being received per day?
- Which gender is spending more time on the screen and on social media?

Having asked these questions, I had to then explore the data structure of my dataset.

Variable	Data Type	Description
Addiction_level	String	Level of Phone addiction
Count	Integer	Number of users
AVG_Screen_Time	Float	Average Screen time
AVG_Social_Media	Float	Average social Media hours
AVG_Gaming_Hrs	Float	Average gaming hours
AVG_WS_Hrs	Float	Average work study hours

From exploring my data types, I had to then identify trends and to see if there were any anomalies;

Screen time vs Addiction level

- Higher addiction levels likely show higher average screen time.

Gaming hours

- Users with high gaming hours had a high chance of falling into higher addiction categories.

Gender vs Average Screen time

- Male individuals were spending more time comparing to Female and Other

Testing Hypothesis

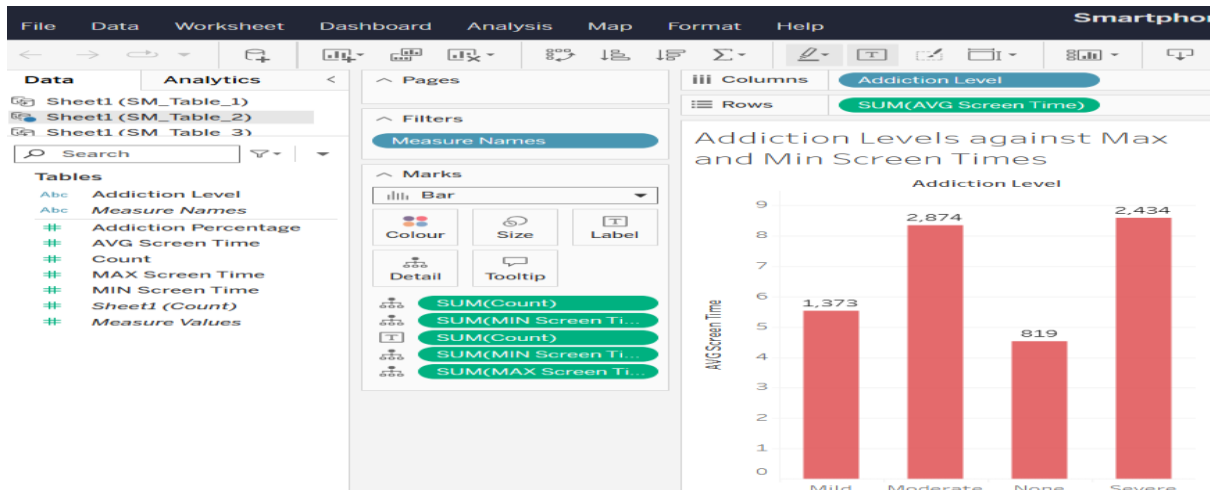
After conducting my performance trends and outliers, I managed to conclude some hypothesis.

1. Users with higher screen time were likely to have higher addiction levels.
2. Social media usage contributes more to addiction than work/study usage
3. Gaming hours increase with addiction level.
4. More male individuals were spending more time on their screens than the other genders

To conclude, the exploratory data analysis revealed a strong relationship between smartphone addiction level and average screen time. Users classified under higher addiction categories showed increased social media and gaming usage. Social media usage appeared to contribute more significantly to addiction patterns compared to work/study usage. A few extreme screen

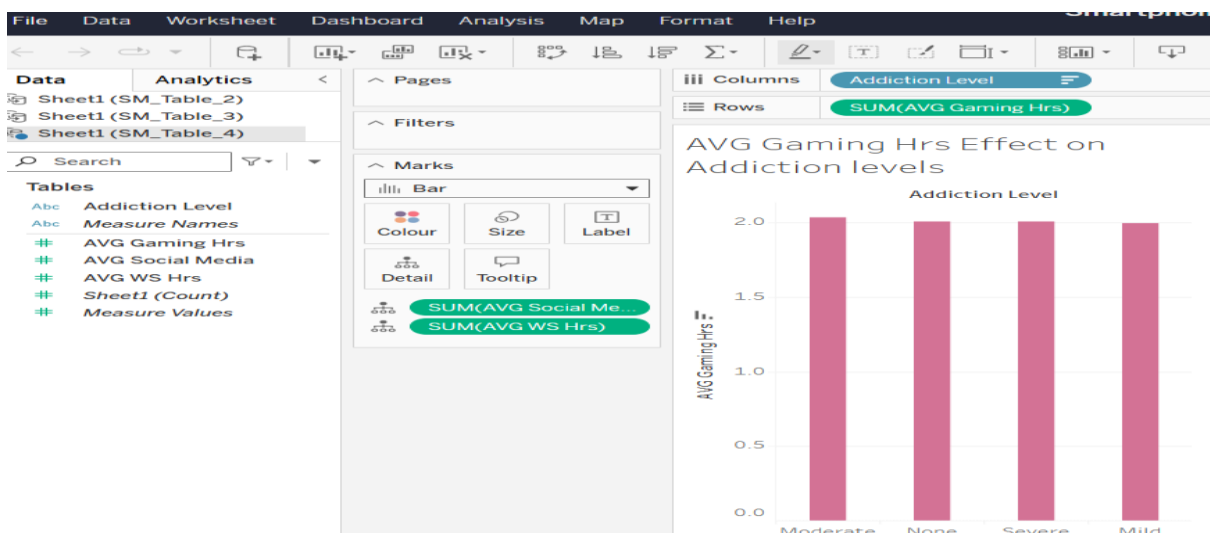
time values were identified as potential outliers and may require further investigation before predictive modelling.

Task 3 – Data Visualization using Tableau



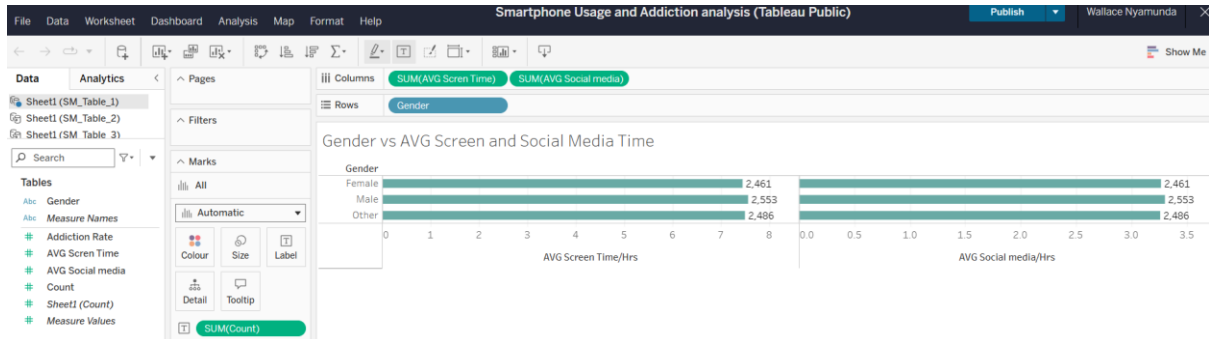
Screen time vs Addiction level

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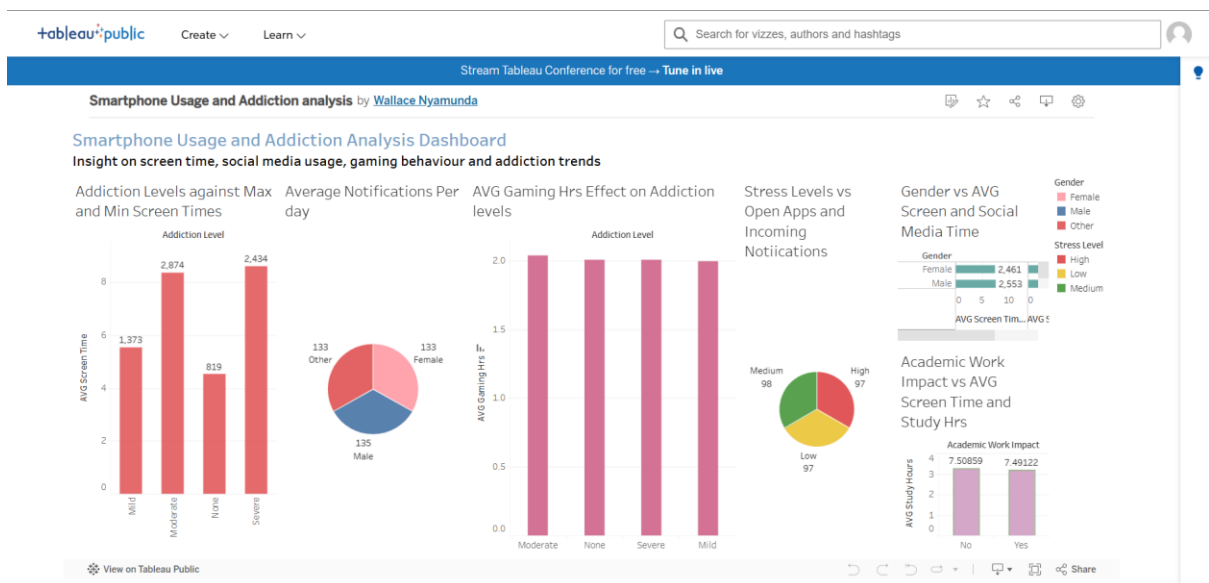
Gaming hours

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Gender vs Average Screen time

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This then shows the dashboard overview of all the trends. Find link to dashboard below:

<https://public.tableau.com/app/profile/wallace.nyamunda/viz/SmartphoneUsageandAddictionanalysis/Dashboard1>

To conclude, the analysis reviewed strong relationship between smartphone addiction and increased screen time, social media activity and gaming behaviour. The Tableau dashboard effectively visualized these trends and gives an interactive way to explore the dataset.